

NAPIER, New Zealand

WMO: 933730

Lat: 39.4703S

Lon: 176.8644E

Elev: 3

StdP: 101.29

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.2	2.4	-1.2	3.4	4.6	-0.3	3.7	5.5	12.5	12.4	11.3	13.1	1.6	240	0.512

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.7	28.3	18.6	26.3	18.0	24.8	17.6	20.6	24.7	19.8	23.6	19.1	22.7	7.2	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.0	22.2	18.5	13.4	21.6	17.7	12.7	21.0	59.5	24.9	56.8	23.7	54.4	22.8	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.8	8.6	-1.2	32.0	0.9	1.2	-1.9	32.8	-2.4	33.5	-2.9	34.1	-3.5	35.0		
(5)			-1.7	21.9	0.9	0.9	-2.4	22.6	-2.9	23.1	-3.4	23.6	-4.0	24.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.0	18.9	18.6	17.1	14.6	12.4	10.0	9.4	9.8	11.5	13.4	15.1	17.6
	DBStd	4.29	2.93	2.67	2.57	2.63	2.80	2.66	2.42	2.32	2.72	2.85	3.01	2.87
	HDD10.0	133	0	0	0	1	8	33	42	32	14	4	1	0
	HDD18.3	1714	28	26	56	117	185	251	278	263	204	155	104	49
	CDD10.0	1597	276	241	219	138	83	32	23	27	60	109	154	235
	CDD18.3	135	45	33	17	4	1	0	0	0	0	2	8	26
	CDH23.3	708	264	170	76	8	1	0	0	0	1	7	43	137
	CDH26.7	133	62	33	8	0	0	0	0	0	0	3	27	27
Wind	WSAvg	3.8	4.1	3.7	3.6	3.3	3.4	3.6	3.7	3.6	3.9	4.3	4.3	4.1
Precipitation	PrecAvg	800	46	59	86	60	68	100	88	79	63	34	50	69
	PrecMax	902	108	124	236	106	243	297	158	183	141	66	103	165
	PrecMin	561	10	4	5	15	12	18	27	22	22	8	6	6
	PrecStd	109	28	40	73	28	57	73	42	53	38	18	27	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.7	30.1	27.7	24.6	22.2	19.3	18.3	18.8	22.0	24.2	26.9	29.5
		MCWB	19.5	19.6	17.9	17.0	16.0	13.3	12.7	12.6	14.4	15.2	17.2	18.7
	2%	DB	28.5	27.4	25.4	22.3	20.2	17.5	16.5	17.1	19.9	22.1	24.5	26.7
		MCWB	18.7	19.1	17.4	16.2	14.6	12.5	11.6	11.6	13.4	14.4	15.8	17.7
	5%	DB	26.2	25.4	23.6	20.7	18.8	16.1	15.1	15.6	18.2	20.5	22.6	24.6
		MCWB	18.1	18.8	17.4	15.8	14.0	11.7	10.8	10.9	12.4	13.9	15.2	17.1
	10%	DB	24.2	23.6	22.2	19.4	17.6	14.9	13.9	14.4	16.7	18.9	20.9	22.9
		MCWB	17.9	18.4	17.1	15.4	13.4	11.3	10.3	10.5	11.9	13.3	14.8	16.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.6	21.9	20.2	18.9	17.2	14.7	13.6	13.6	15.2	16.5	18.4	20.4
		MCDB	26.7	26.1	23.5	21.2	20.4	16.9	16.4	16.5	19.6	21.3	23.2	25.8
	2%	WB	20.6	20.8	19.2	17.7	15.9	13.6	12.5	12.5	14.2	15.5	17.2	19.2
		MCDB	25.1	24.4	22.6	20.0	18.5	15.8	15.0	15.0	18.1	19.9	21.7	23.8
	5%	WB	19.7	20.0	18.5	17.1	15.0	12.7	11.6	11.8	13.4	14.7	16.4	18.4
		MCDB	23.7	23.4	21.7	19.3	17.5	15.1	14.0	14.3	16.7	18.8	20.7	22.5
	10%	WB	18.9	19.3	17.8	16.2	14.1	11.8	10.9	11.1	12.6	14.0	15.6	17.6
		MCDB	22.6	22.6	20.9	18.8	16.6	14.4	13.2	13.8	15.7	17.6	19.8	21.5
Mean Daily Temperature Range	5% DB	MDBR	8.7	8.2	8.7	8.5	8.5	8.3	7.6	8.2	8.8	8.9	8.9	8.4
		MCDBR	12.1	10.9	11.2	10.0	9.8	9.4	9.0	9.8	11.1	11.1	11.4	11.2
		MCWBR	5.6	5.1	5.7	5.6	5.9	6.1	5.6	6.0	6.2	5.9	6.0	5.5
	5% WB	MCDBR	9.1	8.6	8.8	7.4	8.0	8.0	7.3	8.1	9.2	9.3	9.7	8.8
		MCWBR	5.0	4.9	5.4	4.8	5.4	5.7	5.1	5.6	5.9	5.7	5.9	5.0
Clear-Sky Solar Irradiance	taub	0.340	0.339	0.324	0.312	0.294	0.288	0.289	0.296	0.315	0.326	0.329	0.341	
	taud	2.559	2.581	2.614	2.631	2.654	2.639	2.642	2.631	2.548	2.531	2.545	2.554	
	Ebn at Noon	990	967	942	890	848	821	844	896	934	969	996	994	
	Edh at Noon	106	99	88	75	63	59	63	73	92	103	107	107	
All-Sky Solar Radiation	RadAvg	6.63	5.65	4.49	3.14	2.21	1.70	1.90	2.70	3.87	5.13	6.05	6.39	
	RadStd	0.40	0.52	0.30	0.23	0.15	0.12	0.15	0.19	0.31	0.31	0.42	0.45	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	+0.38	+0.61	N/A	N/A	N/A	-30	N/A	N/A	N/A
Regional (5 neighbors)	N/A	+0.40	+0.61	N/A	N/A	N/A	-27	-82	N/A	N/A

Nomenclature: See separate page